

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

June Examination Paper 1 Answer Book

NAME: Marking Guideline

CLASS: 12 _____ TEACHER: _____

Question	Mark			Moderator
	H	T	U	
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
TOTAL (out of 150)				
Percentage				

QUESTION 1

	Solution	Marks
1.1.1	$x(2x - 7) = -4$ $2x^2 - 7x + 4 = 0 \quad \checkmark \text{SF}$ $x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(2)(+4)}}{2(2)} \quad \checkmark$ $\therefore x \approx 2,78 \quad \checkmark \text{both} \quad \text{or} \quad x \approx 0,72$	(3)
1.1.2	$3^{2x-1} + 9^{x-1} - 4 = 0$ $\frac{3^{2x}}{3} + 3^{\overset{\checkmark \wedge}{2x}} \cdot \frac{1}{9} = 4 \quad \checkmark \text{Negative exponents}$ $3 \cdot 3^{2x} + 3^{2x} = 36 \quad \checkmark$ $4 \cdot 3^{2x} = 36$ $3^{2x} = 9 \quad \checkmark$ $3^{2x} = 3^2$ $\therefore 2x = 2$ $x = 1 \quad \checkmark$	(5)

	Solution	Marks
1.1.3	$\sqrt{2x+4} + 5 = x+3$ $\sqrt{2x+4} = x-2$ $(\sqrt{2x+4})^2 = (x-2)^2 \quad \checkmark m$ $2x+4 = x^2 - 4x + 4$ $x^2 - 6x = 0 \quad \checkmark SF$ $x(x-6) = 0 \quad \checkmark \text{Factors}$ $x = 0 \quad \text{or} \quad x = 6 \quad \checkmark$ $N/A \quad \checkmark \text{reject}$	(5)
1.1.4	$2x^2 - 11x \leq 6$ $2x^2 - 11x - 6 \leq 0 \quad \checkmark SF$ $(2x+1)(x-6) \leq 0$ $CV: x = -\frac{1}{2}, x = 6$ $\checkmark CV$  $\therefore -\frac{1}{2} \leq x \leq 6 \quad \checkmark$	(3)

	Solution	Marks
1.2	$2x - y - 1 = 0$ $x^2 - xy = y^2 - 11 \dots \textcircled{2}$ $y = 2x - 1 \dots \textcircled{1} \checkmark$ $x^2 - x(2x - 1) = (2x - 1)^2 - 11 \checkmark_m$ $x^2 - 2x^2 + x = 4x^2 - 4x + 1 - 11$ $5x^2 - 5x - 10 = 0$ $x^2 - x - 2 = 0 \checkmark_{SF}$ $(x - 2)(x + 1) = 0 \checkmark_{\text{factors}}$ $x = 2$ or $x = -1$ $\checkmark_{x\text{-values}}$ $\therefore y = 2(2) - 1$ $\therefore y = 2(-1) - 1$ $y = 3$ $y = -3$ $\checkmark_{y\text{-values}}$	(6)
1.3.1	$x^2 - 8x + 20$ $= x^2 - 8x + (-4)^2 + 20 - (-4)^2$ $= (x - 4)^2 + 4 \checkmark_A > 0 \checkmark_{>0}$ $\therefore A(x)$ is $\mathbb{R}$	(2)
1.3.2	Min under $\sqrt{\quad}$ is 4 $\checkmark$ $\therefore$ min of $A(x)$ is 2 $\checkmark$	(2)
		[26]

QUESTION 2

	Solution	Marks
2.1	$\text{Constant } 2^{\text{nd}} \text{ Diff} = 2a$ $= 2(2)$ $= 4 \quad \checkmark$	(1)
2.2	$3a + b = T_2 - T_1$ $3(2) + b = 2 - (-3) \quad \checkmark$ $\therefore b = -1 \quad \checkmark$ $a + b + c = -3$ $(2) + (-1) + c = -3 \quad \checkmark$ $\therefore c = -4 \quad \checkmark$	(4)
		[5]

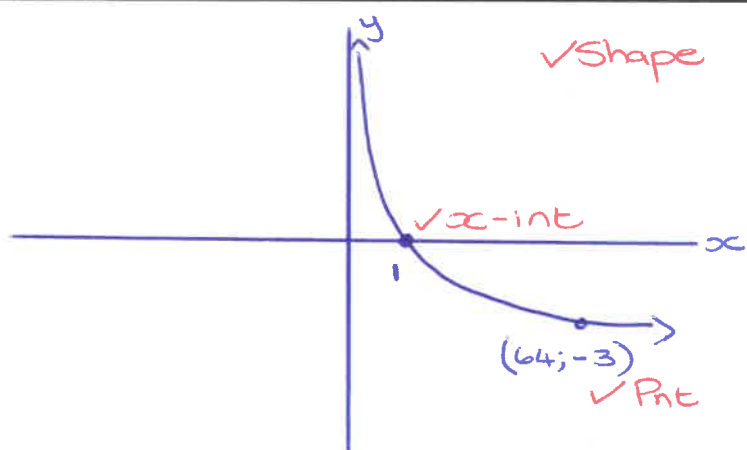
QUESTION 3

	Solution	Marks
3.1.1	$T_n = ar^{n-1}$ $= x^{-3} \cdot (x^{-1})^{n-1} \quad \checkmark a \& r \text{ into correct formula}$ $= x^{-3} \cdot x^{-n+1}$ $= x^{-n-2} \quad \checkmark \text{Single base}$ $\text{Answer only } \checkmark \checkmark$	(2)

	Solution	Marks
3.1.2	$T_{10} = \left(\frac{1}{2}\right)^{-10-2} \checkmark$ $= 4096 \checkmark$	
		(2)
3.2	$\frac{1}{2} - 2 - \frac{9}{2} - \dots - 37$ $a = \frac{1}{2} \quad d = -\frac{5}{2} \checkmark d$ $T_n = -\frac{5}{2}n + 3 \checkmark T_n$ $-\frac{5}{2}n + 3 = -37 \checkmark T_n = -37$ $-\frac{5}{2}n = -40$ $\therefore n = 16 \checkmark$ $\sum_{j=1}^{16} \left(-\frac{5}{2}j + 3\right) \checkmark \text{Sigma Notation}$	
		(5)

	Solution	Marks
3.3	$\frac{a(1-r^4)}{1-r} = 120 \dots \textcircled{1} \quad \frac{a}{1-r} = \frac{243}{2} \dots \textcircled{2}$ $\textcircled{1} \div \textcircled{2} \quad \checkmark_m$ $1-r^4 = \frac{80}{81} \quad \checkmark$ $\therefore r^4 = \frac{1}{81}$ $r = \pm \frac{1}{3} \quad \checkmark \pm \frac{1}{3}$ $\therefore a = 81 \quad \checkmark$	(6)
3.4.1	$S_{100} = -3(100)^2 + 6(100) \quad \checkmark$ $= -29400 \quad \checkmark$ Answer only $\checkmark\checkmark$	(2)
3.4.2	$T_7 = S_7 - S_6 \quad \checkmark_m$ $= -3(7)^2 + 6(7) - [-3(6)^2 + 6(6)] \quad \checkmark$ $= -33 \quad \checkmark$	(3)
		[20]

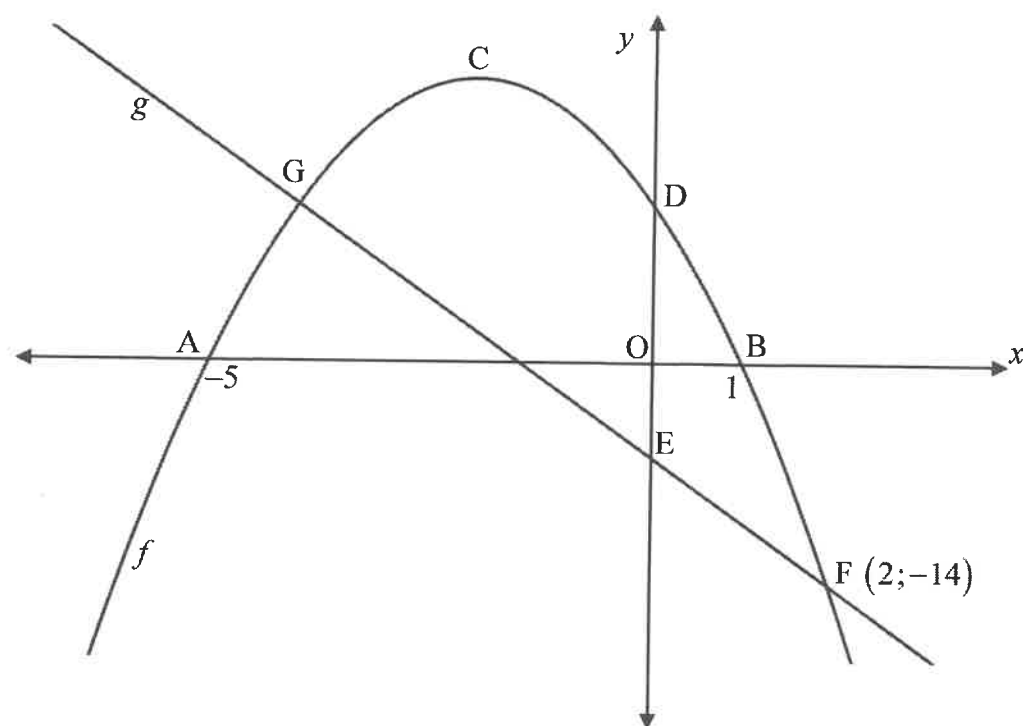
QUESTION 4

	Solution	Marks
4.1	$f(-3) = 4^{-(-3)}$ $= 64 \checkmark$ $(-3; 64)$	(1)
4.2	$f^{-1}: x = 4^{-y}$ $-y = \log_4 x$ $y = -\log_4 x \checkmark$ $\text{Alt: } x = \left(\frac{1}{4}\right)^y \checkmark$ $\therefore y = \log_{\frac{1}{4}} x \checkmark$	(2)
4.3		(3)
4.4	$x > 0 \checkmark$	(1)
4.5	$0 < x \leq 64 \checkmark$	(2)
4.6	$t = -4 \checkmark$	(1)
		[10]

QUESTION 5

	Solution	Marks
5.1	$x = 3$ ✓	(1)
5.2	$y = \frac{x-5}{x-3}$ $= \frac{x-3-2}{x-3}$ $= \frac{x-3}{x-3} - \frac{2}{x-3}$ $y = \frac{-2}{x-3} + 1$ ✓	(2)
5.3	✓ Shape & orientation ✓ Horizontal as. $y=1$ ✓ $x=3$ Vert As ✓ $x=5$ x-int ✓ y-int $\frac{3}{5}$	(5)
5.4	$y = -x + 4$ ✓	(2)
		[10]

QUESTION 6



	Solution	Marks
6.1	$y = a(x+5)(x-1) \checkmark$ $-14 = a(2+5)(2-1) \checkmark$ $-14 = 7a$ $\therefore a = -2 \checkmark$ $y = -2(x+5)(x-1)$ $= -2(x^2 + 4x - 5)$ $= -2x^2 - 8x + 10$ $\therefore a = -2; b = -8; c = 10$	(3)

	Solution	Marks
6.2	$x = \frac{-(-8)}{2(-2)} = -2 \checkmark$ $f(-2) = 18 \checkmark \therefore C(-2; 18)$ $\therefore \text{Range}_f : y \leq 18 \checkmark$	(3)
6.3	$D(0; 10) \text{ both } \checkmark E(0; -6)$ $\therefore DE = 16 \text{ units } \checkmark$ $\text{Answer only } \checkmark$	(2)
6.4	$-5 \leq x \leq 1 \checkmark \text{ All or nothing}$	(2)
6.5	$f'(x) = -4x - 8 \checkmark f'$ $-4x - 8 = -4 \checkmark f' = -4$ $-4x = 4$ $x = -1 \checkmark$ $f(-1) = 16 \therefore \text{Contact point } (-1; 16) \checkmark$ $h(x): y - 16 = -4(x + 1)$ $y = -4x + 12$ $\therefore K = 12 \checkmark$	(5)
6.6	$K > 12 \checkmark$	(1)
		[16]

	Additional Space	Marks
6.5	Alt:	
	$h(x) = -4x + k$ $f' = m$	
	$-4x + k = -2x^2 - 8x + 10$ $\checkmark h = f$	
	$2x^2 + 4x + k - 10 = 0$	
	$\Delta = (4)^2 - 4(2)(k - 10)$	
	$= -8k + 96$ $\checkmark \Delta$	
	Equal roots: $\Delta = 0$	
	$-8k + 96 = 0$ $\checkmark \Delta = 0$	
	$-8k = -96$	
	$\therefore k = 12$ $\checkmark$	

QUESTION 7

	Solution	Marks
7.1	$A = P(1-i)^n$ $5690 = 10000(1-i)^{3,5} \quad \checkmark \text{subs into}$ $i = 0,14879... \quad \checkmark \text{correct formula}$ $\therefore r = 14,88\% \checkmark$	(3)
7.2	$F = \frac{\infty [(1+i)^n - 1]}{i}$ $= \frac{9000 \left[\left(1 + \frac{7,5\%}{12}\right)^{60} - 1 \right]}{\frac{7,5\%}{12}} \quad \checkmark \text{subs into}$ $= 65\,2743,95 \checkmark \quad \text{correct formula}$ $\therefore A = 652743,95 \left(1 + \frac{7,5\%}{12}\right)^1 \quad \checkmark \text{Growth 1 month}$ $= \underline{R\,65\,6823,60} \checkmark$	(5)

	Solution	Marks
7.3	$P = \frac{x [1 - (1+i)^{-n}]}{i}$ ✓ subs into correct formula $4\,500\,000 = \frac{50\,000 [1 - (1 + \frac{10\%}{12})^{-n}]}{\frac{10\%}{12}}$ $\left(\frac{121}{120}\right)^{-n} = \frac{1}{4} \quad \checkmark$ $\therefore -n = \log_{\frac{121}{120}} \left(\frac{1}{4}\right) \quad \checkmark \text{ use of log}$ $n = 167,0475... \quad \checkmark$ $\therefore 168 \text{ withdrawals } \checkmark$	(6)
		[14]

QUESTION 8

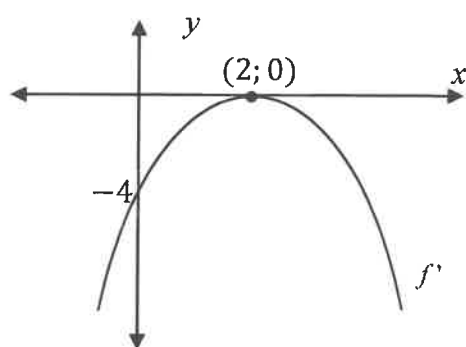
	Solution	Marks
8.1	$f(x+h) = -(x+h)^3$ $= -(x+h)(x^2 + 2hx + h^2)$ $= -(x^3 + 3x^2h + 3xh^2 + h^3)$ $= -x^3 - 3x^2h - 3xh^2 - h^3$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \checkmark$ $= \lim_{h \rightarrow 0} \frac{-x^3 - 3x^2h - 3xh^2 - h^3 - (-x^3)}{h} \quad \checkmark f(x+h)$ $= \lim_{h \rightarrow 0} \frac{-3x^2h - 3xh^2 - h^3}{h}$ $= \lim_{h \rightarrow 0} \frac{h(-3x^2 - 3xh - h^2)}{h} \quad \checkmark \text{HCF}$ $= \lim_{h \rightarrow 0} (-3x^2 - 3xh - h^2)$ $= -3x^2 - 3x(0) - (0)^2 \quad \checkmark \text{Apply limit}$ $= -3x^2 \quad \checkmark$	

(5)

	Solution	Marks
8.2	$D_x \left[\frac{7}{x} - \sqrt[3]{x} \right]$ $D_x \left[7x^{-1} - x^{\frac{1}{3}} \right] \checkmark$ $= -7x^{-2} - \frac{1}{3}x^{-\frac{2}{3}}$ $\checkmark A \quad \quad 3 \quad \checkmark A$	(3)
8.3.1	$y = kx^2 + k$ $\frac{dy}{dx} = 2kx \checkmark$	(1)
8.3.2	$y = k(x^2 + 1) \checkmark$ $k = \frac{y}{x^2 + 1} \quad \checkmark \text{ In terms of } k$ $\frac{dk}{dy} = \frac{1}{x^2 + 1} \checkmark$	(3)

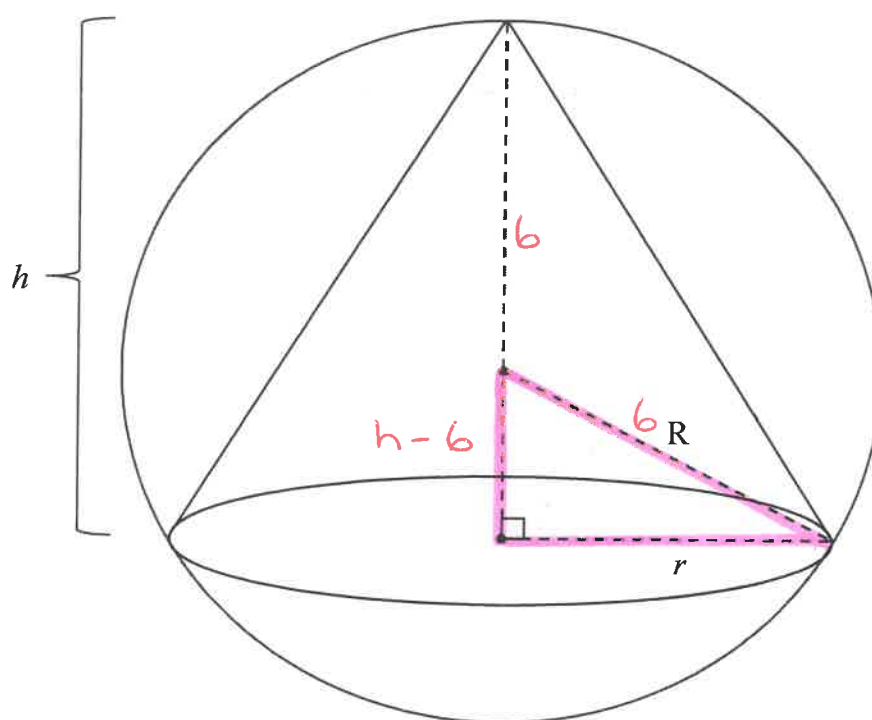
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QUESTION 9



	Solution	Marks
9.1	$x \in \mathbb{R} \checkmark$	(1)
9.2	$x = 2 \checkmark$	(1)
9.3	Concave up $\checkmark$	(1)
9.4	$x = 0 \checkmark$; $x = 4 \checkmark$	(2)
9.5	Hand-drawn graph of a parabola opening downwards on a Cartesian coordinate system. The vertex is marked with a dot at the point $(2; 9)$. Red handwritten notes to the right of the graph state: $\checkmark$Shape, $\checkmark$Orientation, and $\checkmark$SPOI. The x and y axes are labeled with arrows.	
		(3)
		[8]

QUESTION 10



	Solution	Marks
10.1	$r^2 = (6)^2 - (h-6)^2 \quad \checkmark m$ $r^2 = 36 - (h^2 - 12h + 36)$ $r^2 = 12h - h^2 \quad \checkmark$ $V = \frac{1}{3} \pi r^2 h$ $= \frac{1}{3} \pi (12h - h^2) h \quad \checkmark$ $= \frac{\pi}{3} (12h^2 - h^3)$	
		(4)

	Solution	Marks
10.2	$V = 4\pi h^2 - \frac{\pi h^3}{3}$ $V' = 8\pi h - \pi h^2 \quad \checkmark V'$ $8\pi h - \pi h^2 = 0 \quad \checkmark V' = 0$ $\pi h(8 - h) = 0 \quad \checkmark \text{Factors}$ $\therefore h = 0 \quad \text{or} \quad h = 8 \quad \checkmark \text{Answers}$ with selection N/A $\text{Max } V = 4\pi(8)^2 - \frac{\pi(8)^3}{3}$ $\approx 268,08 \text{ cm}^3 \quad \checkmark$	(5)
		[9]

QUESTION 11

	Solution	Marks
11.1	$n(\text{flashcards}) = 30$ $\therefore \text{Total learners} = 60 \checkmark$	(1)
11.2	$x = 30 - 2$ $= 28 \checkmark \checkmark$	(2)
11.3	$P(S \text{ and } F) = \frac{2}{60} \checkmark = \frac{1}{30}$	(1)
11.4	$P(\text{FC}) = 0,5$ $P(\text{Pass}) = \frac{52}{60} \checkmark = \frac{13}{15} \quad P(\text{Pass})$ $P(\text{FC and Pass}) = \frac{24}{60} \checkmark = \frac{2}{5} \quad P(\text{FC and Pass})$ $P(\text{FC}) \times P(\text{Pass}) = 0,5 \times \frac{13}{15}$ $= \frac{13}{30}$ $\therefore P(\text{FC and Pass}) \neq P(\text{FC}) \times P(\text{Pass}) \checkmark \text{ Conclusion}$ $\therefore \text{Not independent}$	(5)
		[9]

QUESTION 12

	Solution	Marks
12.1	$9 \cdot 8 \cdot 7 \cdot 6 \checkmark$	(2)
	$= 3024 \checkmark$	
12.2	$8 \cdot 7 \cdot 6 \cdot 4 \checkmark$	(2)
	$= 1344 \checkmark$	
12.3	$\frac{7 \cdot 6 \cdot 3 \cdot 2}{3024} \quad \checkmark 7 \times 6 \times 3 \quad \checkmark \times 2$	(4)
	$\frac{1}{12} \quad \checkmark \text{Answer}$	
		[8]